Maulana Azad National Institute of Technology, Bhopal

Mid Term Examination (November 18, 2024)

Session: July to December 2024

Course: B.Tech. (Sem I) **Section:** All

Subject: Mathematics - I **Subject Code:** MTH 24101

Duration: 60 Minutes **Maximum Marks:** 20

Instructions:
- All questions are compulsory and carry equal marks.
- Calculator is not allowed.
- Proper justifications should be given for each answer.

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1. If $x = r\cos\theta$, $y = r\sin\theta$, prove that $\dfrac{\partial^2 r}{\partial x^2} + \dfrac{\partial^2 r}{\partial y^2} = \dfrac{1}{r}\left\{\left(\dfrac{\partial r}{\partial x}\right)^2 + \left(\dfrac{\partial r}{\partial y}\right)^2\right\}$.

2. Find the extreme values of $f(x, y, z) = x^2 + y^2 + z^2 + xy + xz + yz$ subject to the constraints $x + y + z = 1$ and $x + 2y + 3z = 3$.

3. Trace the curve $y^2(1 - x^2) = x^2(1 + x^2)$ by describing the appropriate properties.

4. Find the area bounded by the curves in the first quadrant $y = x^2$, $8x = y^2$ and $y = (\sqrt{3} - 1)(2x + \sqrt{3})$.

** End of the question paper **

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Maulana Azad National Institute of Technology, Bhopal

Mid Term Examination (February, 2022) [Online Assessment]

Course: B. Tech. **Semester-I** **Branch: All**
Subject: Mathematics-I **Sub. Code: MTH-111**
Time: 90 Minutes **Date: 04-02-2022** **Max. Marks: 20**

Instructions: There are five questions. Attempt all questions. Your writing should be readable. Assume missing data if any. Please capture the images of the handwritten solutions and make a single PDF, where the **filename** should be **your scholar number** and upload the file in response to your solution. Please write **your name and scholar number** on each page of the response sheet.

S.No	Question	Marks
1	Solve the following integral $\int_0^{\infty}\int_{-\infty}^{\infty} 5\, e^{-(x^2+y^2)}\, dxdy$	04
2	Trace the curve for the equation $y(x^2 - a^2) = (x^2 + a^2)$ after calculating the following: i) Check the concavity and coordinates of inflection points of f(x). ii) Identify symmetry and asymptotes	04
3	Find the Tayler Series around (1, 1) of $f(x, y) = \tan^{-1}(xy)$, and then compute the value of *f (1.1, 0.8)* using Taylor Series quadratic approximation.	04
4	If the temperature T at any point (x, y, z) in space is T = 4 xyz^2, then using Lagrange's Multiplier method, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.	04
5	If u = $\sin^{-1}\left(\frac{x^6+y^6}{x+y}\right)$, then calculate $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$.	04

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